EXPERIMENT - 12

16 Bit division.

AIM:

To write an assembly language program to implement 16 bit divided using 8086 processor.

ALGORITHM:

1) Read dividend (16) bit.
2) Read divisor.
3) count <- 8.
4) left shift dividend.
5) Subtract divisor from upper 8 bits of dividend.
6) if cs = 1, go to 9.
7) Restore dividend.
8) Increment lower 8 bit of dividend
9) count <- count - 1
10) if count = 0, go to 5.
11) store upper 8 bit dividend as remainder and lower 8 bit as quotient.
12) stop.

PROGRAM:

MNEMONICS	EXPLANATION
MOV AX, [1100H]	MOVE A to 1100.

MOV BX, 1102H	Move B to 1102_2
DIV BX	Divide by base (B)
MOV [1200H], AX	Move [1200] to A
MOV [1202H], DX	Move [1202] to direction (D)
HLT	Halt.

Input

Address	Data
1100	10
1102	10

Output :

Address	Data
1200	1
1202	10

RESULT:

Thus the program was executed successfully using 8086 processor simulator.